Koi McFarland

Computer Science Engineer · Self Taught Programmer · AI Trainer

Charleston, SC, USA

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Summary

Passionate and self-taught software developer and engineer with a proven work ethic, demonstrated through success in fast-paced environments and highly demanding academic settings. Earned a degree in Computer Science with a minor in Mathematics from Charleston Southern University, graduating with a GPA above 3.90. Proficient in C++, Java, and Python, with additional experience in C, C#, Rust, JavaScript, and AI development. Skilled in both object-oriented and procedural programming, with hands-on knowledge of full-stack development, database management, and Linux systems. Experienced in game and simulation development using Unity, Godot, and Unreal Engine 5, as well as Git/GitHub version control. Strong background in collaborative problem-solving, demonstrated through ICPC/CCPC programming competitions and complex team-based academic projects.

Education

Charleston Southern University

North Charleston, SC

B.S. in Computer Science and Engineering; Minor in Mathematics

Class of 2026

- Member, Alpha Chi National College Honor Society
- CSU Jairy C. Hunter Scholarship Recipient
- SC LIFE Scholarship Recipient
- Dean's List: 2023, 2024, 2025
- GPA: 3.9/4.0

Internships

NIWC / Core4ce

North Charleston, SC

ACE Cyber R&D Internship

Aug. 2025 - Dec. 2025

- Conducted research and authored a whitepaper for NIWC.
- Designed and implemented a conversational AI interface for querying Elasticsearch databases.
- Presented research findings at NIWC facilities to technical staff and program stakeholders.
- For references, contact Prof. Henderson (`jhenderson@csuniv.edu`) or Prof. Littleton (`wlittleton@csuniv.edu`).

Research & Publication

OntologyWeb

Live Demo

Project Description

- Research prototype for visualizing, curating, comparing, and editing ontology and triple-based knowledge structures.
- Supports RDF/OWL ontology visualization and LLM-assisted generation of compact triples and ontology drafts from software requirements text.
- Enables side-by-side structural comparison of baseline and current requirement graphs, with visual encoding of additions, deletions, and modifications.

Technical Description

- Built in Python with Dash and dash-cytoscape for the interactive graph UI; RDF/OWL handling with rdflib.
- Integrates a local LLM via Ollama for chunked triple and Turtle ontology generation, with evidence provenance tracing back to source text.
- Features snapshot-based version storage, semantic-layer filtering, and graph JSON/RDF export.
- Hosted on Render with the source code maintained in a GitHub repository.

Publication & Conference

- Co-authored the paper *"LLM-Assisted Workflow for Structural Difference Visualization in Evolving Software Requirements"* with Songhui Yue.
- Accepted as a Poster & Tool Demo at the 2026 IEEE 34th International Requirements Engineering Conference (RE 2026).
- IEEE DOI: 10.1109/RE68928.2026.00070; preprint available on arXiv: arXiv:2609.28002.

GitHub Repository

- Public repository — github.com/songhui01/OntologyWeb.

Programming Competitions

2024	**1st Place - CofC D2**, ICPC NA SOUTH DIVISION	*Charleston, SC*
2024	**Top 10 - Cofc D2**, ICPC NA SOUTH DIVISION	*Charleston, SC*
2024	**Top 10 - FU**, CCSC SE REGION	*Greenville, SC*
2023	**Top 20 - CCU**, CCSC SE REGION	*Conway, SC*

Current Employment

DataAnnotation

New York, NY (Remote)

Software Developer - AI Trainer (Contract)

Apr. 2024 - Present

- Trains AI's coding abilities by examining, scrutinizing, and rewriting their code.
- Grades the AI's code based on a list of questions and criteria given by the company.
- Write a comprehensive report on how the AI performed while writing code.

Personal Projects

True Size Of Games

www.truesizeofgames.com

Project Description

- Displays the scale of popular video game maps compared to real-world geography.

Technical Description

- Built as a static site using JavaScript, HTML, and CSS, with the Leaflet library powering the interactive world map.
- Fully optimized as a Progressive Web App (PWA) with top scores in SEO, accessibility, and performance.
- Hosted on Netlify with the source code maintained in a GitHub repository.

GitHub Repository

- Private repository — access available upon request.

Falling Sand Simulation (Cellular Automata)

github.com/koi-mcfarland/Falling-Sand-Simulation

Project Description

- Real-time cellular automata particle simulation running on a 384×216 grid.
- Features multiple materials — sand, dirt, coal, salt, stone, wood, water, oil, smoke, and steam.
- Materials interact with realistic and varied reactions, such as salt dissolving in water and smoke rising.

Technical Description

- Two development branches:
  - Main branch — Object-Oriented Programming (OOP) approach.
  - ECS branch — Entity Component System approach using EnTT. (currently not functional)
- Programmed entirely in C++ using SDL2, with hardware-accelerated texture rendering and fixed-timestep physics updates.
- Built with Make; fully cross-platform (Windows, macOS, Linux).

GitHub Repository

- Public repository — available on my GitHub.